Living organisms that require oxygen to respire can build up hydrogen peroxide ($H_2O_2$) as a byproduct of respiration. Because hydrogen peroxide can cause oxidative damage to cells, the amount of $H_2O_2$ in cells must be closely regulated. Although $H_2O_2$ decomposes spontaneously, a catalyst is needed for the reaction to occur at a rate sufficiently rapid for biological purposes. In biological systems, an enzyme called a catalase accelerates the catalytic decomposition of $H_2O_2$ into water and oxygen (Reaction 1).

$$2\,H_2O_2(aq) \xrightarrow{\text{catalyst}} 2\,H_2O(l) + O_2(g) + \text{heat}$$

**Reaction 1**

In a laboratory setting, several different inorganic compounds can also serve as catalysts for $H_2O_2$ decomposition, allowing the comparison of how different catalysts affect the efficiency of the reaction. To compare $Fe^{3+}$ (a metal homogeneous catalyst) and $I^-$ (a halogen homogeneous catalyst), researchers measured the amount of heat evolved after placing salts of each catalyst in a fresh solution of 3% hydrogen peroxide.

To perform the measurements, researchers placed 50 mL of 3% $H_2O_2$ in an insulated coffee cup to be used as a calorimeter, as shown in Figure 1. Before adding any catalyst, the initial (baseline) temperature of the $H_2O_2$ solution was determined by recording the temperature every 30 seconds for 5 minutes. Then 10 mL of 0.10 M $Fe(NO_3)_3(aq)$, a source of the $Fe^{3+}$ catalyst, was added to the $H_2O_2$ solution, and the temperature was recorded every 30 seconds for another 15 minutes. The experiment was then repeated using 10 mL of 0.50 M $NaI(aq)$ (dissolved in a basic solution) as a source of the $I^-$ catalyst. The results for each reaction are shown in Figure 2.

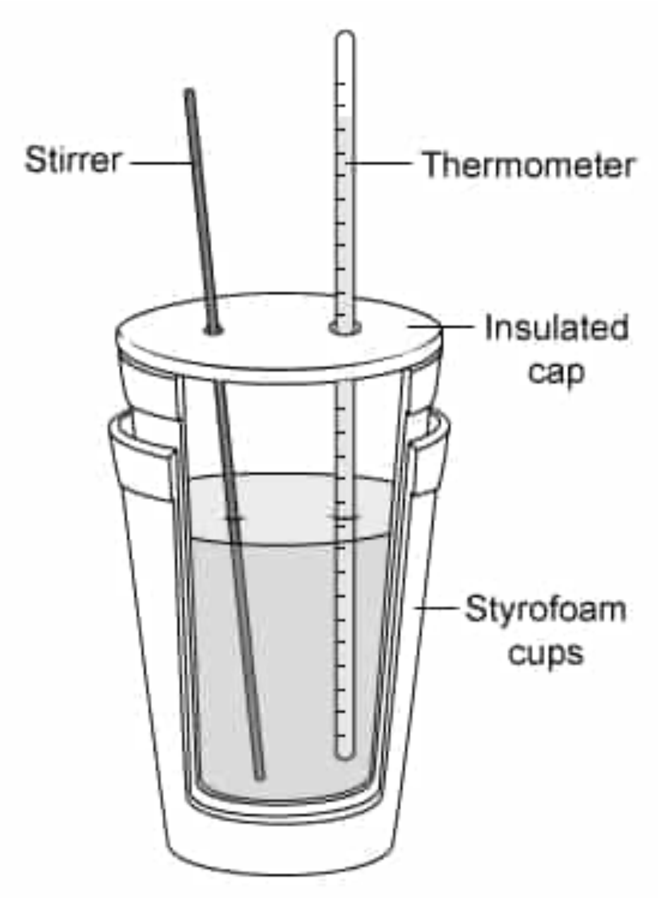



**Figure 1** Coffee cup calorimeter

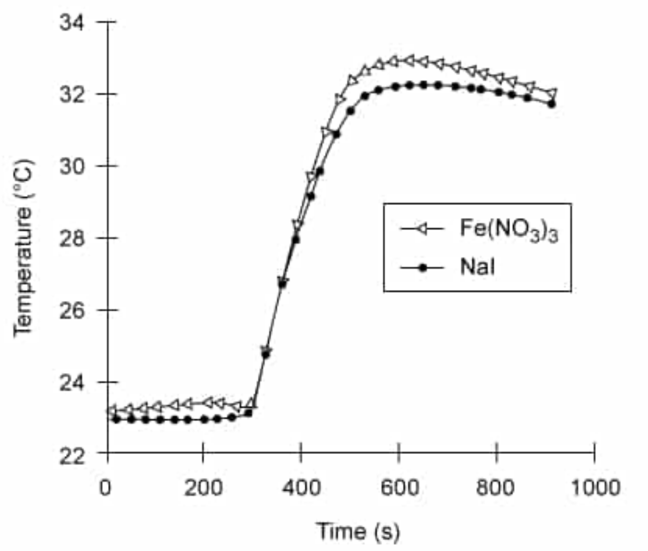


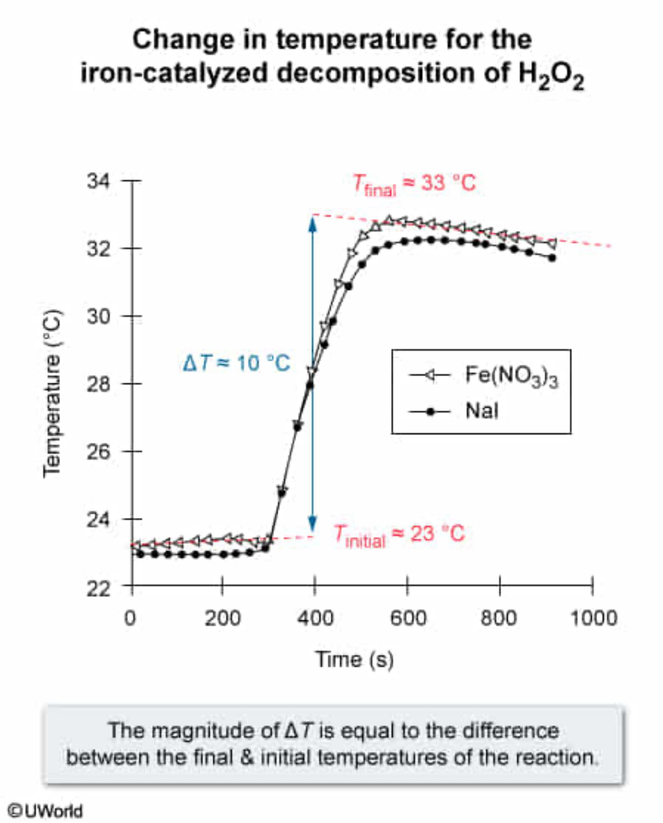

**Figure 2** Temperature vs. time measurements for reactions catalyzed by $Fe^{3+}$ and $I^-$

Based on the results in Figure 2, $\Delta T$ for the decomposition of $H_2O_2$ using $Fe(NO_3)_3$ is closest to which of the following?

- A. −11 °C [2%]
- B. 10 °C [86%]
- C. 23 °C [5%]
- D. 33 °C [6%]

Correct
85% answered correctly

**Explanation:**




A **calorimetry** experiment measures the **temperature** ***T*** as a reaction progresses. The results are displayed graphically with temperature (the dependent variable) on the *y*-axis and time (the independent variable) on the *x*-axis. Any **change in temperature Δ*T*** caused by heat that is either released or absorbed during the reaction is seen as a sharp increase or decrease along the *y*-axis. $\Delta T$ is found by subtracting the **initial temperature** $T_i$ at the beginning of the reaction from the **final temperature** $T_f$ at the end of the reaction.

$$\Delta T = T_f - T_i$$

Although $T_f$ and $T_i$ are usually evaluated using trend lines, $\Delta T$ can be reasonably estimated by using temperatures immediately before and after the sharp vertical change in the graph as the values of $T_f$ and $T_i$.

The experiment described in the passage observed an increase in temperature caused by the heat evolved from the catalytic decomposition of $H_2O_2$. The approximate $\Delta T$ for the reaction can be found by evaluating $T_f$ and $T_i$. Figure 2 shows that the temperature at the beginning of the reaction ($T_i$) is 23 °C. According to Reaction 1, heat is one of the products of the reaction; therefore, when the reaction stops producing heat (ie, the temperature stops rising), the reaction has ended. On this basis, the temperature at the end of the reaction ($T_f$) is approximately 33 °C.

Substituting these values of $T_f$ and $T_i$ into the equation above gives:

$$\Delta T = T_f - T_i = 33\ ℃ - 23\ ℃$$

$$\Delta T = 10\ ℃$$

**(Choice A)** A temperature of −11 °C is found if the $I^-$ catalyzed reaction is evaluated instead of the $Fe^{3+}$ catalyzed reaction, and the $T_i$ and $T_f$ are switched. Heat is released from the reaction, making $T_f$ higher than $T_i$; therefore, $\Delta T$ should be positive.

**(Choices C and D)** Temperatures of 23 °C and 33 °C are the approximate initial and final temperatures, respectively, rather than the *change* in temperature.

### Educational objective:

Calorimetry experiments measure the temperature as a reaction progresses. Any change in temperature $\Delta T$ during the reaction is found by subtracting the initial temperature $T_i$ at the beginning of the reaction from the final temperature $T_f$ at the end of the reaction.

**Time spent:**0 sec
**Subject:**
General Chemistry

**QID**:402342
**System:**
Thermodynamics, Kinetics & Gas Laws

Living organisms that require oxygen to respire can build up hydrogen peroxide ($H_2O_2$) as a byproduct of respiration. Because hydrogen peroxide can cause oxidative damage to cells, the amount of $H_2O_2$ in cells must be closely regulated. Although $H_2O_2$ decomposes spontaneously, a catalyst is needed for the reaction to occur at a rate sufficiently rapid for biological purposes. In biological systems, an enzyme called a catalase accelerates the catalytic decomposition of $H_2O_2$ into water and oxygen (Reaction 1).

$$2\,\mathrm{H_2O_2}(aq) \xrightarrow{\text{catalyst}} 2\,\mathrm{H_2O}(l) + \mathrm{O_2}(g) + \text{heat}$$

**Reaction 1**

In a laboratory setting, several different inorganic compounds can also serve as catalysts for $H_2O_2$ decomposition, allowing the comparison of how different catalysts affect the efficiency of the reaction. To compare $Fe^{3+}$ (a metal homogeneous catalyst) and $I^-$ (a halogen homogeneous catalyst), researchers measured the amount of heat evolved after placing salts of each catalyst in a fresh solution of 3% hydrogen peroxide.

To perform the measurements, researchers placed 50 mL of 3% $H_2O_2$ in an insulated coffee cup to be used as a calorimeter, as shown in Figure 1. Before adding any catalyst, the initial (baseline) temperature of the $H_2O_2$ solution was determined by recording the temperature every 30 seconds for 5 minutes. Then 10 mL of 0.10 M $Fe(NO_3)_3(aq)$, a source of the $Fe^{3+}$ catalyst, was added to the $H_2O_2$ solution, and the temperature was recorded every 30 seconds for another 15 minutes. The experiment was then repeated using 10 mL of 0.50 M NaI(*aq*) (dissolved in a basic solution) as a source of the $I^-$ catalyst. The results for each reaction are shown in Figure 2.

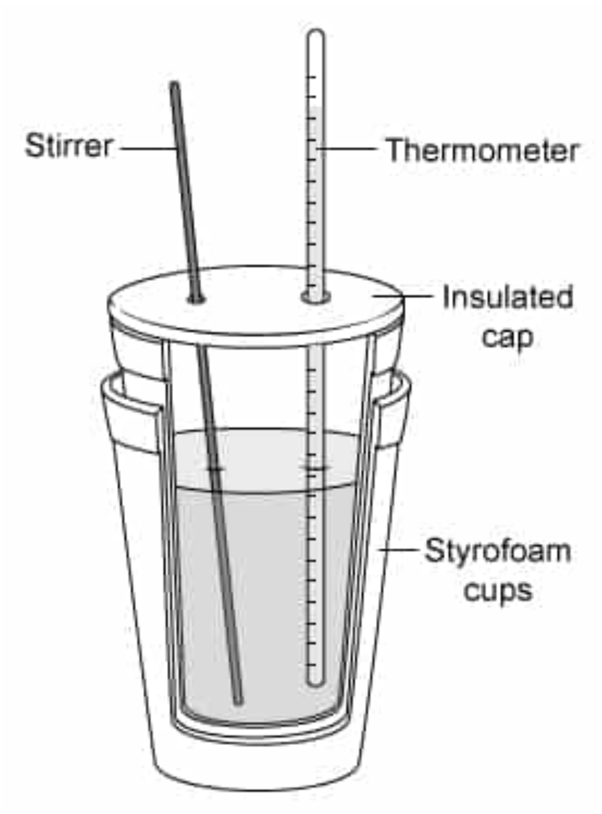



**Figure 1** Coffee cup calorimeter

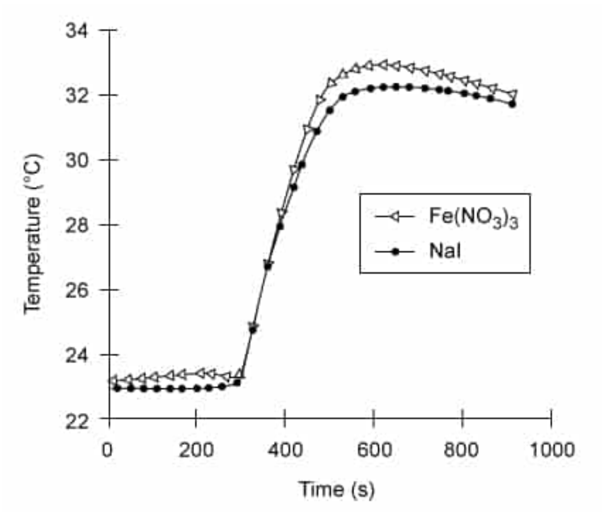
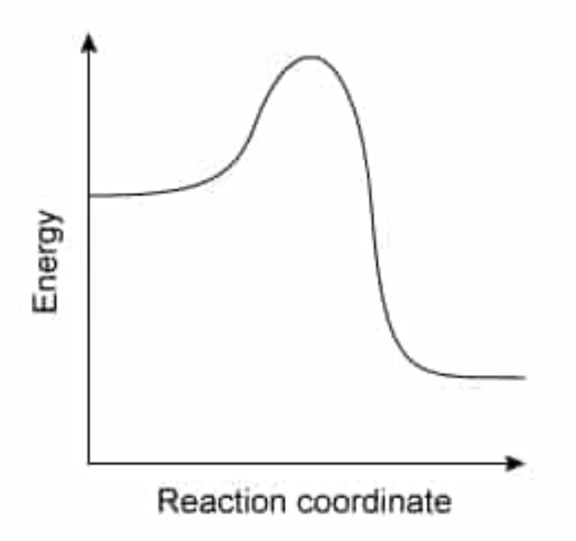



**Figure 2** Temperature vs. time measurements for reactions catalyzed by $Fe^{3+}$ and $I^{-}$

Below is a reaction energy diagram for the decomposition of $H_2O_2$.

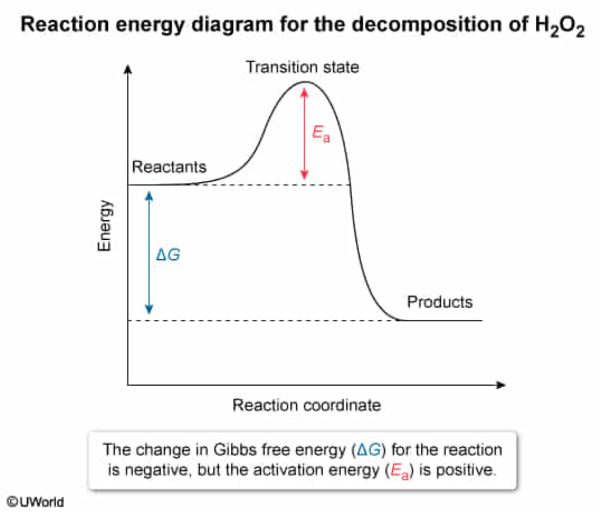



As the reaction proceeds, the change in the Gibbs free energy of the reaction is:

- A. negative, but the activation energy is positive. [95%]
- B. equal to the activation energy.
- C. positive, but the activation energy is negative. [2%]
- D. zero, but the activation energy is nonzero and positive.

Correct
96% answered correctly

**Explanation:**




The energy changes that occur during the progress of a reaction are depicted by a **reaction energy diagram** that plots the **Gibbs free energy** *G* (*y*-axis) versus the reaction coordinate (*x*-axis), which indicates the reaction progress. The energy difference between the reactants and the products is the change in Gibbs free energy $\Delta G$:

$$\Delta G = G_{\text{products}} - G_{\text{reactants}}$$

If the reactants are higher in energy than the products, Δ$G$ is negative, which indicates that the reaction is spontaneous. Conversely, if the reactants are lower in energy than the products, Δ$G$ is positive, which indicates that the reaction is nonspontaneous.

The activation energy $E_a$ (ie, the minimum amount of energy required to initiate a reaction) is the difference between the free energy of the transition state and the energy of the reactants. Because $E_a$ is the energy required by the reactants to form a higher energy transition state, $E_a$ is **always positive**.

As seen in the reaction energy diagram for the decomposition of $H_2O_2$ given in this question, the reactants are higher in energy than the products; consequently, Δ$G$ is negative. Therefore, the decomposition of $H_2O_2$ proceeds with a negative Δ$G$ and a positive $E_a$.

**(Choices B, C, and D)** According to the reaction energy diagram for the decomposition of $H_2O_2$, Δ$G$ is a *negative* value because the products are lower in energy than the reactants, and $E_a$ is a *positive* value because the transition state is higher in energy than the reactants. As a result, Δ$G$ and $E_a$ cannot be equal, and Δ$G$ cannot be zero because the reactants and products have different energies.

**Educational objective:**
A reaction coordinate diagram represents the Gibbs free energy during the progress of a reaction. The change in Gibbs free energy Δ$G$ for a reaction is determined by subtracting the free energy of the reactants from that of the products. The difference in energy between the transition state and the reactants is the activation energy, which is always a positive value.

**Time spent:**0 sec
**Subject:**
General Chemistry

**QID**:402343
**System:**
Thermodynamics, Kinetics & Gas Laws

Living organisms that require oxygen to respire can build up hydrogen peroxide ($H_2O_2$) as a byproduct of respiration. Because hydrogen peroxide can cause oxidative damage to cells, the amount of $H_2O_2$ in cells must be closely regulated. Although $H_2O_2$ decomposes spontaneously, a catalyst is needed for the reaction to occur at a rate sufficiently rapid for biological purposes. In biological systems, an enzyme called a catalase accelerates the catalytic decomposition of $H_2O_2$ into water and oxygen (Reaction 1).

$$2\,H_2O_2(aq) \xrightarrow{\text{catalyst}} 2\,H_2O(l) + O_2(g) + \text{heat}$$

**Reaction 1**

In a laboratory setting, several different inorganic compounds can also serve as catalysts for $H_2O_2$ decomposition, allowing the comparison of how different catalysts affect the efficiency of the reaction. To compare $Fe^{3+}$ (a metal homogeneous catalyst) and $I^-$ (a halogen homogeneous catalyst), researchers measured the amount of heat evolved after placing salts of each catalyst in a fresh solution of 3% hydrogen peroxide.

To perform the measurements, researchers placed 50 mL of 3% $H_2O_2$ in an insulated coffee cup to be used as a calorimeter, as shown in Figure 1. Before adding any catalyst, the initial (baseline) temperature of the $H_2O_2$ solution was determined by recording the temperature every 30 seconds for 5 minutes. Then 10 mL of 0.10 M $Fe(NO_3)_3(aq)$, a source of the $Fe^{3+}$ catalyst, was added to the $H_2O_2$ solution, and the temperature was recorded every 30 seconds for another 15 minutes. The experiment was then repeated using 10 mL of 0.50 M NaI(*aq*) (dissolved in a basic solution) as a source of the $I^-$ catalyst. The results for each reaction are shown in Figure 2.

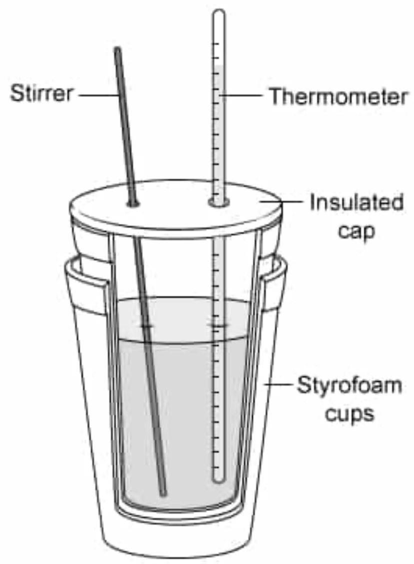



**Figure 1** Coffee cup calorimeter

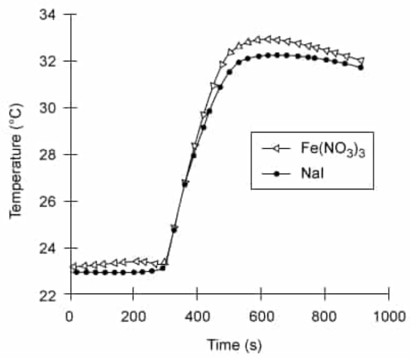



**Figure 2** Temperature vs. time measurements for reactions catalyzed by $Fe^{3+}$ and $I^-$

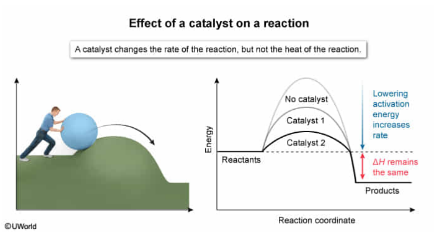
Which of the following statements explains why the change in temperature of the $H_2O_2$ decomposition reaction using

$Fe^{3+}$ as a catalyst is approximately the same value as the change in temperature using $I^-$ as a catalyst?

- A. $Fe^{3+}$ stabilizes the transition state more than $I^-$. [3%]
- B. $Fe^{3+}$ results in a faster reaction rate than $I^-$. [2%]
- C. $Fe^{3+}$ and $I^-$ both increase the total amount of products produced. [14%]
- D. $Fe^{3+}$ and $I^-$ do not affect the heat of the reaction. [80%]

Correct
79% answered correctly

**Explanation:**




In contrast to the thermodynamics of a reaction—which indicate whether a reaction is spontaneous or not—the kinetics of a reaction describe how long a reaction takes to reach completion under given conditions. A reaction may be spontaneous but may also proceed so slowly that it does not reach completion in a reasonable amount of time. A **catalyst** increases the **rate** (kinetics) of the reaction by decreasing the activation energy required for the reaction to progress at a given temperature. However, a catalyst does not affect the amounts of the products produced or the enthalpy (thermodynamics) of the reaction.

The decomposition of $H_2O_2$ is an exothermic reaction. No matter how long the decomposition takes, the reaction produces a certain amount of heat per mole of reactants; the identity of the catalyst does not affect the heat of the reaction. Therefore, the reaction using $Fe^{3+}$ as the catalyst results in the same change in temperature as the reaction using $I^-$ as a catalyst, assuming the same amount of $H_2O_2$ is used.

**(Choice A)** Although a catalyst stabilizes the high-energy transition state of a reaction, the extent to which a catalyst stabilizes the transition state does not affect the heat evolved from the reaction.

**(Choice B)** The reaction rate achieved by a catalyst has no bearing on the heat evolved from the reaction.

**(Choice C)** A catalyst affects only the *rate* at which the products are produced, not the total amount of products produced.

**Educational objective:**
A catalyst increases the rate of a reaction by lowering the activation energy needed for the reaction to occur. A catalyst does not change the amounts of the products produced or the enthalpy (heat) of the reaction.

**Time spent:** 0 sec
**Subject:** General Chemistry

**QID:** 402344
**System:** Thermodynamics, Kinetics & Gas Laws

Living organisms that require oxygen to respire can build up hydrogen peroxide ($H_2O_2$) as a byproduct of respiration. Because hydrogen peroxide can cause oxidative damage to cells, the amount of $H_2O_2$ in cells must be closely regulated. Although $H_2O_2$ decomposes spontaneously, a catalyst is needed for the reaction to occur at a rate sufficiently rapid for biological purposes. In biological systems, an enzyme called a catalase accelerates the catalytic decomposition of $H_2O_2$ into water and oxygen (Reaction 1).

$$2\,H_2O_2(aq) \xrightarrow{\text{catalyst}} 2\,H_2O(l) + O_2(g) + \text{heat}$$

**Reaction 1**

In a laboratory setting, several different inorganic compounds can also serve as catalysts for $H_2O_2$ decomposition, allowing the comparison of how different catalysts affect the efficiency of the reaction. To compare $Fe^{3+}$ (a metal homogeneous catalyst) and $I^-$ (a halogen homogeneous catalyst), researchers measured the amount of heat evolved after placing salts of each catalyst in a fresh solution of 3% hydrogen peroxide.

To perform the measurements, researchers placed 50 mL of 3% $H_2O_2$ in an insulated coffee cup to be used as a calorimeter, as shown in Figure 1. Before adding any catalyst, the initial (baseline) temperature of the $H_2O_2$ solution was determined by recording the temperature every 30 seconds for 5 minutes. Then 10 mL of 0.10 M $Fe(NO_3)_3(aq)$, a source of the $Fe^{3+}$ catalyst, was added to the $H_2O_2$ solution, and the temperature was recorded every 30 seconds for another 15 minutes. The experiment was then repeated using 10 mL of 0.50 M NaI($aq$) (dissolved in a basic solution) as a source of the $I^-$ catalyst. The results for each reaction are shown in Figure 2.

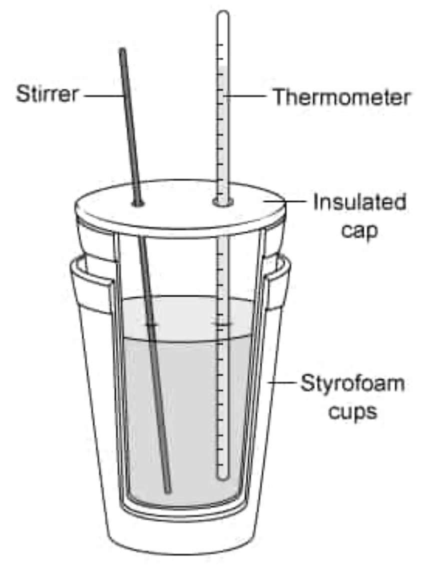



**Figure 1** Coffee cup calorimeter

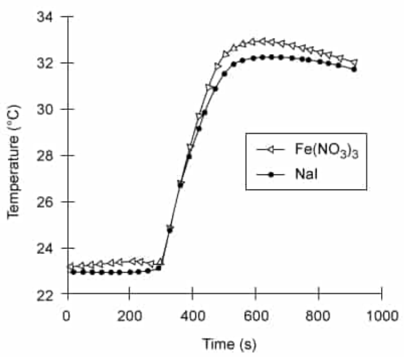


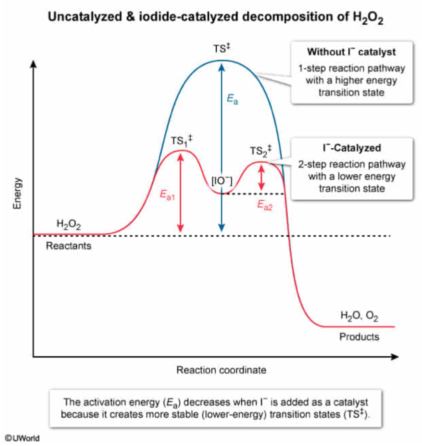

**Figure 2** Temperature vs. time measurements for reactions catalyzed by $Fe^{3+}$ and $I^-$

How is the activation energy for the decomposition of $H_2O_2$ affected by the addition of $I^-$ to the coffee cup?

- ◯ A. Increased, because adding $I^-$ changes the concentration of the reactants in the reaction vessel [1%]
- ◯ B. Increased, because $I^-$ lowers the bond dissociation energy of $H_2O_2$ [2%]
- ✔ ◉ C. Decreased, because $I^-$ stabilizes the transition state of the reaction [93%]
- ◯ D. Decreased, because $I^-$ increases the amount of heat produced by the reaction [1%]

Correct
92% answered correctly

**Explanation:**




A catalyst is a compound that increases the rate of a reaction without being consumed by the reaction. It accomplishes this by lowering the reaction's activation energy $E_a$, the difference between the energy of the transition state and the energy of the reactants. The **transition state** is an unstable, transient formation (activated complex) that is much higher in energy than the reactants. A catalyst **stabilizes** the transition state for a reaction, creating a less energetic transition state. After the products are formed, the catalyst is recovered.

The iodide ion ($I^-$) from NaI acts as a catalyst in the decomposition of $H_2O_2$ by stabilizing the transition state formed when the bonds in $H_2O_2$ break to form $H_2O$ and $O_2$. This decreases the activation energy and increases the rate of decomposition of $H_2O_2$.

**(Choice A)** In the presence of $I^-$, the activation energy is *decreased*, not increased. In addition, $I^-$ does not change the amounts of reactants or products, only the *rate* at which products form.

**(Choice B)** Although the decomposition of $H_2O_2$ requires overcoming the bond dissociation energy of the H–O and O–O bonds, a catalyst such as $I^-$ only serves to stabilize the transition state, causing the bonds to break more quickly. A catalyst does not change the inherent energy required to break the bonds.

**(Choice D)** The activation decreases in the presence of $I^-$, but this decrease is not due to changes in the amount of heat produced by the reaction. The activation energy affects only the kinetics, not the enthalpy of the reaction.

**Educational objective:**
The activation energy of a reaction is the difference in energy between the reactants and the transition state of a reaction. Catalysts lower the activation energy by stabilizing the high-energy transition state.

| | |
|---|---|
| **Time spent:** 0 sec | **QID:** 402345 |
| **Subject:** General Chemistry | **System:** Thermodynamics, Kinetics & Gas Laws |

Living organisms that require oxygen to respire can build up hydrogen peroxide ($H_2O_2$) as a byproduct of respiration. Because hydrogen peroxide can cause oxidative damage to cells, the amount of $H_2O_2$ in cells must be closely regulated. Although $H_2O_2$ decomposes spontaneously, a catalyst is needed for the reaction to occur at a rate sufficiently rapid for biological purposes. In biological systems, an enzyme called a catalase accelerates the catalytic decomposition of $H_2O_2$ into water and oxygen (Reaction 1).

$$2\,\mathrm{H_2O_2}(aq) \xrightarrow{\text{catalyst}} 2\,\mathrm{H_2O}(l) + \mathrm{O_2}(g) + \text{heat}$$

**Reaction 1**

In a laboratory setting, several different inorganic compounds can also serve as catalysts for $H_2O_2$ decomposition, allowing the comparison of how different catalysts affect the efficiency of the reaction. To compare $Fe^{3+}$ (a metal homogeneous catalyst) and $I^-$ (a halogen homogeneous catalyst), researchers measured the amount of heat evolved after placing salts of each catalyst in a fresh solution of 3% hydrogen peroxide.

To perform the measurements, researchers placed 50 mL of 3% $H_2O_2$ in an insulated coffee cup to be used as a calorimeter, as shown in Figure 1. Before adding any catalyst, the initial (baseline) temperature of the $H_2O_2$ solution was determined by recording the temperature every 30 seconds for 5 minutes. Then 10 mL of 0.10 M $Fe(NO_3)_3(aq)$, a source of the $Fe^{3+}$ catalyst, was added to the $H_2O_2$ solution, and the temperature was recorded every 30 seconds for another 15 minutes. The experiment was then repeated using 10 mL of 0.50 M NaI($aq$) (dissolved in a basic solution) as a source of the $I^-$ catalyst. The results for each reaction are shown in Figure 2.

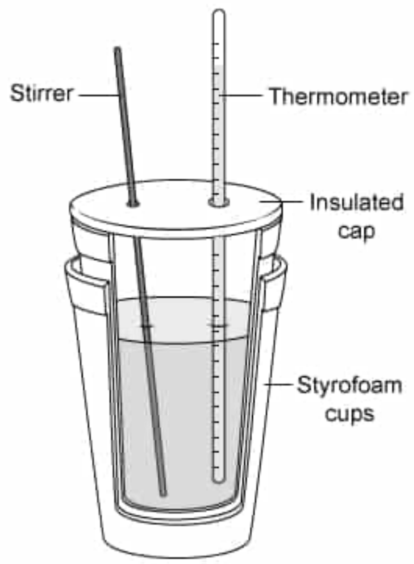



**Figure 1** Coffee cup calorimeter

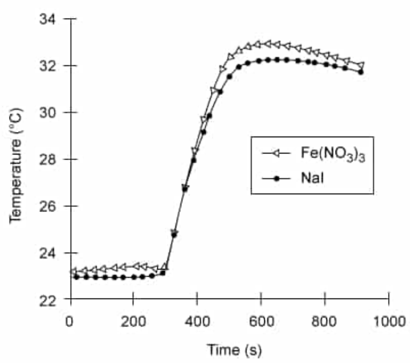



**Figure 2** Temperature vs. time measurements for reactions catalyzed by $Fe^{3+}$ and $I^-$

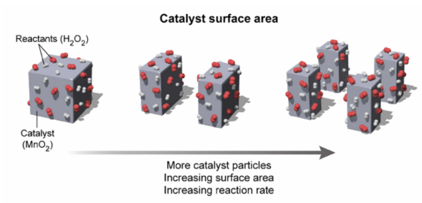
The experiment was repeated with 3.0 g of solid $MnO_2$ granules added as an alternative catalyst. When the $MnO_2$

granules were pulverized to a fine powder, the temperature of the solution rose more quickly than when the granules were not pulverized. The most likely reason for this is that with heterogeneous catalysts:

- A. the rate increases when the catalyst is oxidized. [3%]
- ✔ B. the rate increases as the catalyst surface area is increased. [89%]
- C. the rate decreases due to competing side reactions. [2%]
- D. the rate decreases when the catalyst is insoluble. [4%]

Correct
90% answered correctly

**Explanation:**




Catalysts can be either homogenous or heterogeneous. A **homogenous catalyst** is in the same phase as the reactants whereas a **heterogeneous catalyst** is in a different phase than the reactants. As such, heterogeneous catalysts are usually solids interacting with reactants that are either liquids or gases.

In contrast to absorption, in which one species is incorporated into another medium, a surface (ie, heterogeneous) catalyst works by **adsorption**, in which the reacting species clings to the surface of the catalytic medium. This adherence promotes the interaction of two reactants or weakens the bonds in the species. **Increasing the surface area** of a catalyst **increases the rate** of the reaction because a greater surface area allows more molecules to collect on the surface.

For the system in this question, the $MnO_2$ granules serve as a heterogeneous solid catalyst for the decomposition of $H_2O_2$. The question states that the temperature increased more quickly (ie, the reaction rate was greater) when the catalyst granules were pulverized to a fine powder. The smaller particles of $MnO_2$ provided more surface area to be exposed to $H_2O_2$ molecules, which allowed the reaction to proceed at a faster rate.

**(Choice A)** Using a fine powder instead of larger granules is a physical change to the catalyst, not a chemical change. The chemical (redox) properties of the catalyst do not explain why the physical change in the catalyst influences the reaction rate.

**(Choice C)** In this case, there is only one reactant and no major competing side reactions that would affect the rate of the reaction.

**(Choice D)** The solubility of a homogeneous catalyst can affect the rate of the reaction because higher solubility can increase the interactions between reactant and catalyst species in solution. However, this question involves a heterogeneous catalyst, so the surface area is more important than solubility.

**Educational objective:**
A heterogeneous catalyst is in a different phase than the reactants whereas a homogeneous catalyst is in the same phase as the reactants. Heterogeneous catalysts involve the adsorption of reactants onto the surface of the catalyst and provide higher reaction rates with increased surface area.

**Time spent:**0 sec
**Subject:**
General Chemistry

**QID:**402346
**System:**
Thermodynamics, Kinetics & Gas Laws

Living organisms that require oxygen to respire can build up hydrogen peroxide ($H_2O_2$) as a byproduct of respiration. Because hydrogen peroxide can cause oxidative damage to cells, the amount of $H_2O_2$ in cells must be closely regulated. Although $H_2O_2$ decomposes spontaneously, a catalyst is needed for the reaction to occur at a rate sufficiently rapid for biological purposes. In biological systems, an enzyme called a catalase accelerates the catalytic decomposition of $H_2O_2$ into water and oxygen (Reaction 1).

$$2\,H_2O_2(aq) \xrightarrow{\text{catalyst}} 2\,H_2O(l) + O_2(g) + \text{heat}$$

**Reaction 1**

In a laboratory setting, several different inorganic compounds can also serve as catalysts for $H_2O_2$ decomposition, allowing the comparison of how different catalysts affect the efficiency of the reaction. To compare $Fe^{3+}$ (a metal homogeneous catalyst) and $I^-$ (a halogen homogeneous catalyst), researchers measured the amount of heat evolved after placing salts of each catalyst in a fresh solution of 3% hydrogen peroxide.

To perform the measurements, researchers placed 50 mL of 3% $H_2O_2$ in an insulated coffee cup to be used as a calorimeter, as shown in Figure 1. Before adding any catalyst, the initial (baseline) temperature of the $H_2O_2$ solution was determined by recording the temperature every 30 seconds for 5 minutes. Then 10 mL of 0.10 M $Fe(NO_3)_3(aq)$, a source of the $Fe^{3+}$ catalyst, was added to the $H_2O_2$ solution, and the temperature was recorded every 30 seconds for another 15 minutes. The experiment was then repeated using 10 mL of 0.50 M NaI(*aq*) (dissolved in a basic solution) as a source of the $I^-$ catalyst. The results for each reaction are shown in Figure 2.

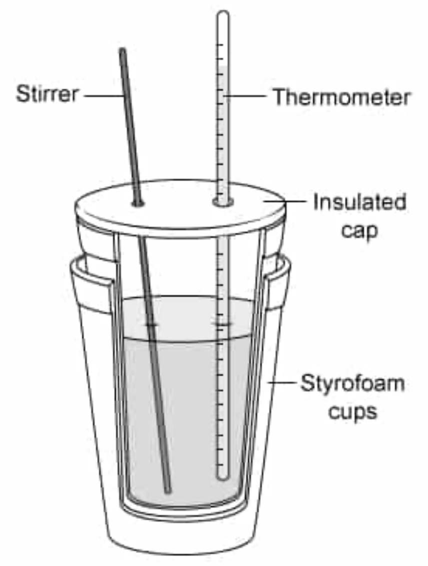



**Figure 1** Coffee cup calorimeter

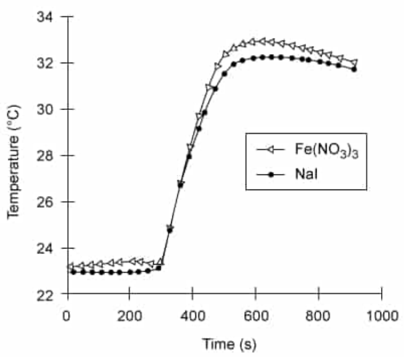



**Figure 2** Temperature vs. time measurements for reactions catalyzed by $Fe^{3+}$ and $I^-$

The results shown in Figure 2 would most likely NOT be affected by which of the following changes to the experiment?

○ A. Adding $AgNO_3(aq)$ to precipitate $I^-$ from the solution as $AgI(s)$ [3%]

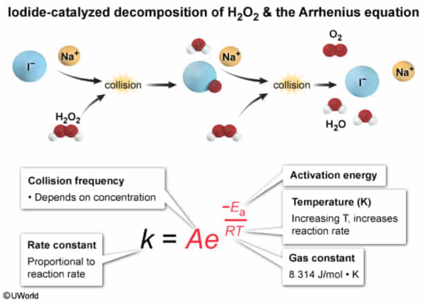
○ B. Using 30% $H_2O_2$ [6%]

✔ ● C. Replacing 0.50 M $NaI(aq)$ with 0.50 M $KI(aq)$ [87%]

○ D. Heating the reaction to an initial temperature of 35 °C [3%]

Correct
86% answered correctly

**Explanation:**




One way to think about kinetics is the **collision theory**, which asserts that molecules must collide to react. Moreover, these collisions must have sufficient kinetic energy to overcome the activation energy $E_a$ of the reaction. This concept is described mathematically using the **Arrhenius equation**:

$$k = Ae^{\frac{-E_a}{RT}}$$

where $k$ is the rate constant of the reaction and $A$ is a constant that represents the collision frequency and structural factors specific to the reactants. The exponential term $e^{\frac{-E_a}{RT}}$ corresponds to the fraction of collisions with enough kinetic energy to overcome the activation energy $E_a$ at a given temperature $T$. $R$ is the gas constant (8.314 $J \cdot mol^{-1} \cdot K^{-1}$).

In the iodide-catalyzed decomposition of $H_2O_2$, $I^-$ ions must collide with $H_2O_2$ for the catalyzed reaction to occur. Changing the catalyst's cation from $Na^+$ to $K^+$ does not change the concentration of $I^-$ or the concentration of $H_2O_2$, and therefore does not change the frequency of collisions in the reaction. Consequently, the data in Figure 2 will remain the same because changing the catalyst cation does not affect the rate at which heat evolves from the reaction (ie, the slope of the line remains unchanged).

**(Choice A)** Precipitating $I^-$ ions (ie, the catalyst) from the solution as $AgI(s)$ will decrease the rate of the reaction because little to no catalyst remains, therefore changing the results in Figure 2.

**(Choice B)** Increasing the concentration of $H_2O_2$ from 3% to 30% will increase the frequency of the collisions $A$ in the reaction, which increases the rate of the reaction and the amount of heat produced.

**(Choice D)** Increasing the initial temperature of the reaction $T$ will result in a greater number of molecular collisions with an energy greater than $E_a$. This increases the rate of the reaction and changes the value of $\Delta T$ because the initial temperature is changed.

**Educational objective:**
The collision theory of kinetics assumes that molecules must collide for a reaction to occur. The collisions must have enough kinetic energy to overcome the activation energy. The reaction rate can be described using the Arrhenius equation.

**Time spent:** 0 sec
**Subject:**
General Chemistry

**QID:** 402347
**System:**
Thermodynamics, Kinetics & Gas Laws